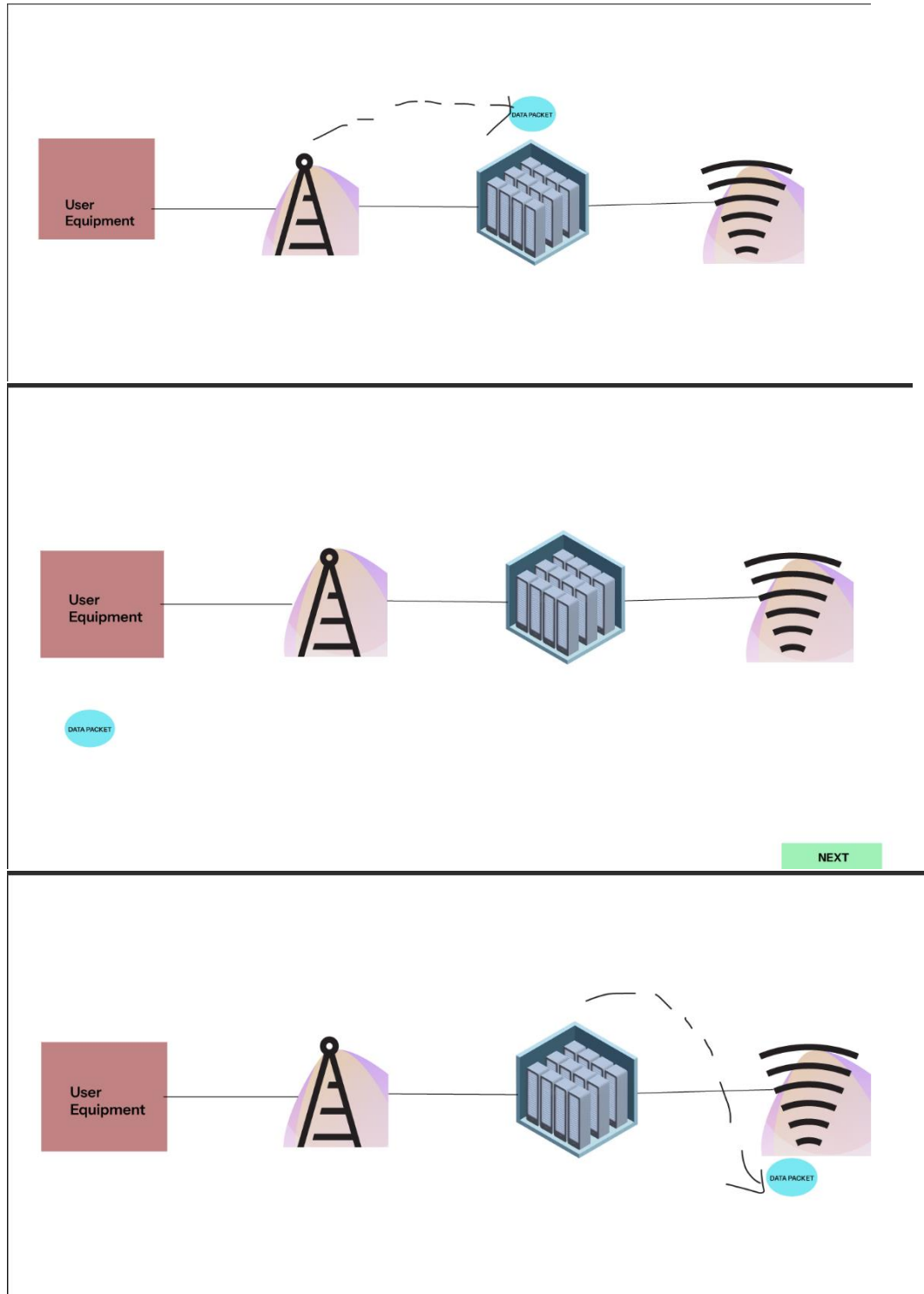
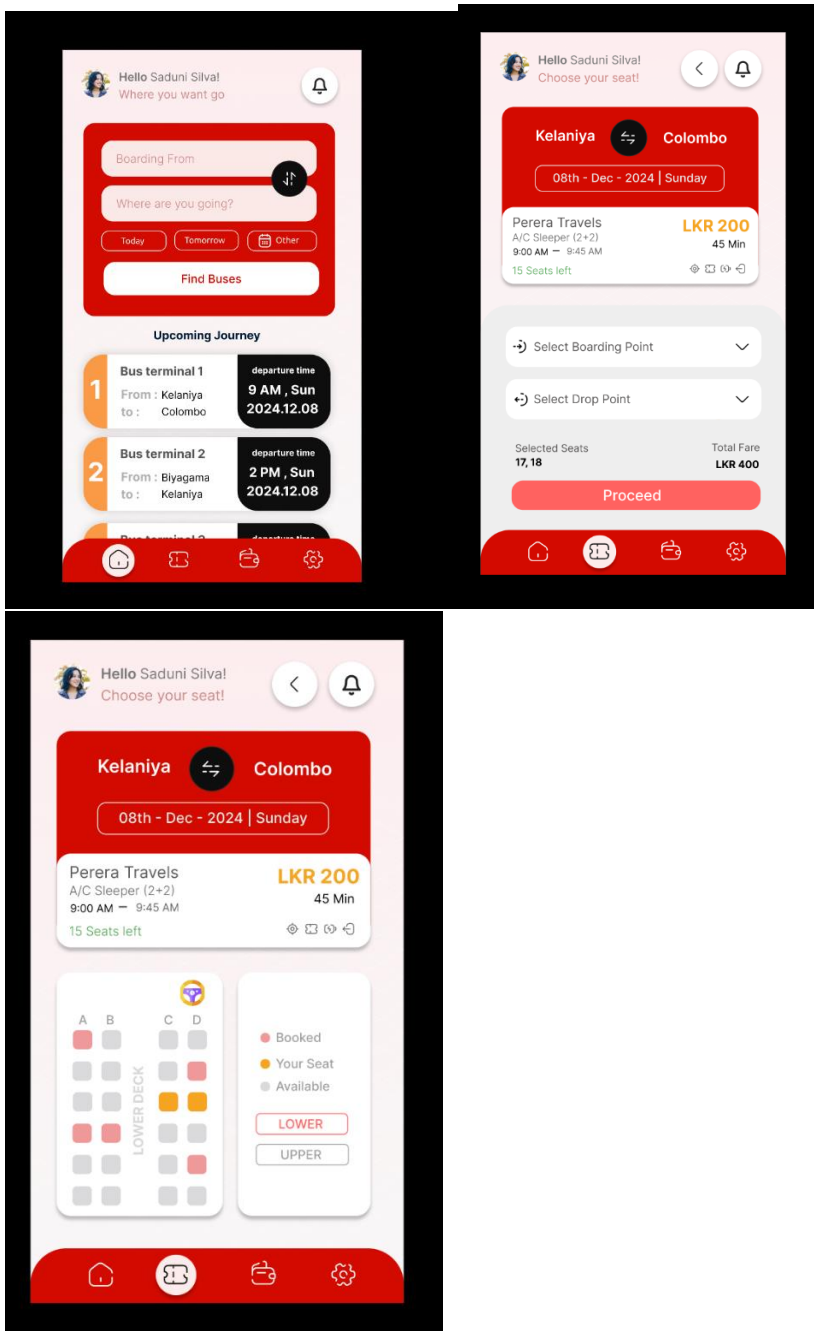
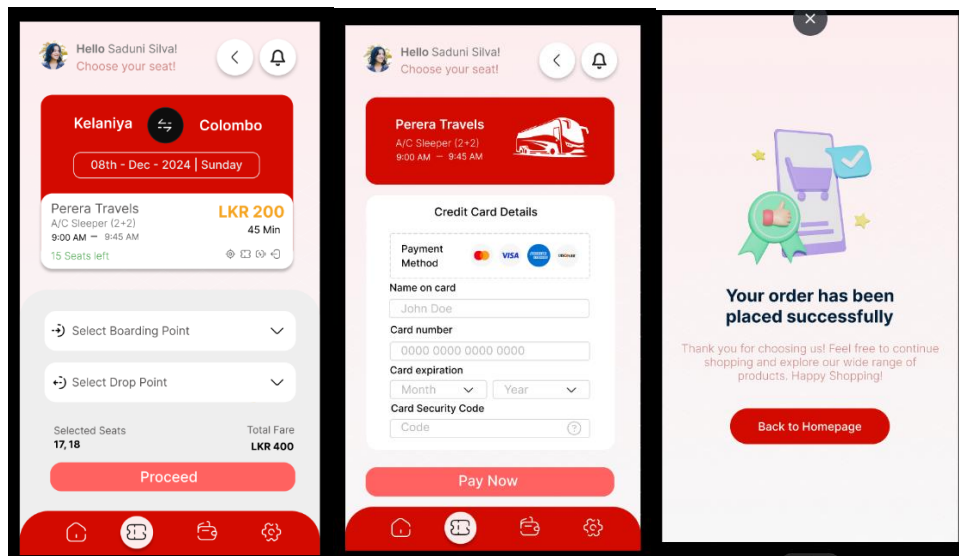


1. <https://www.figma.com/proto/sEIA0JeXI1msGqJhybrdye/Free-Networking-icons--Community-?node-id=202-60&m=draw&scaling=min-zoom&content-scaling=fixed&page-id=0%3A180&starting-point-node-id=202%3A12&show-protocol-sidebar=1&t=GG42kTP7qB90JyyQ-9>



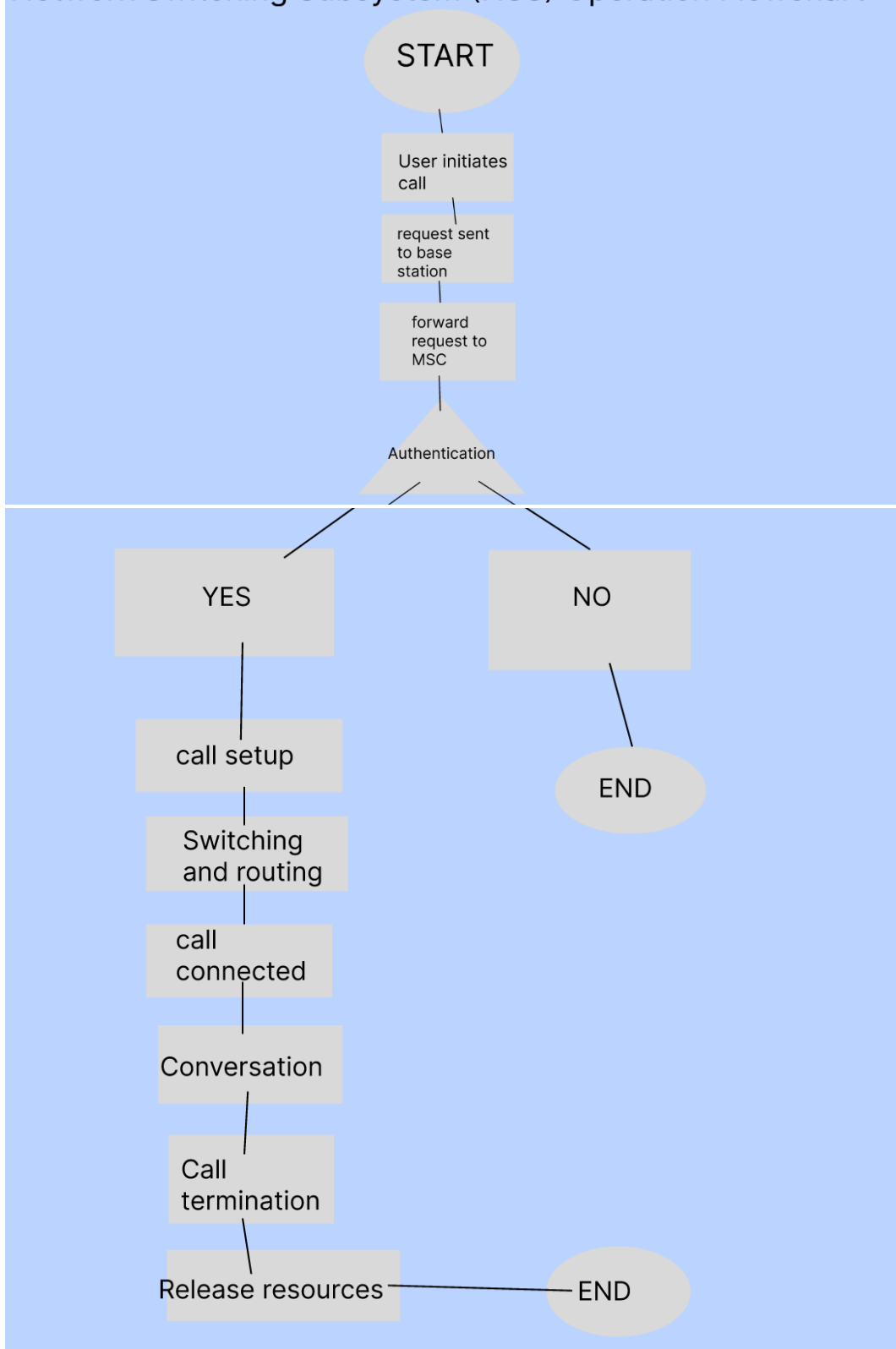
2. <https://www.figma.com/proto/t4YRJovVN3ChJf3KzPmy3L/Online-Bus-Seat-Booking-App--Community-?node-id=4-827&p=f&t=sYCy2qIIZK6SNcl3-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A4>



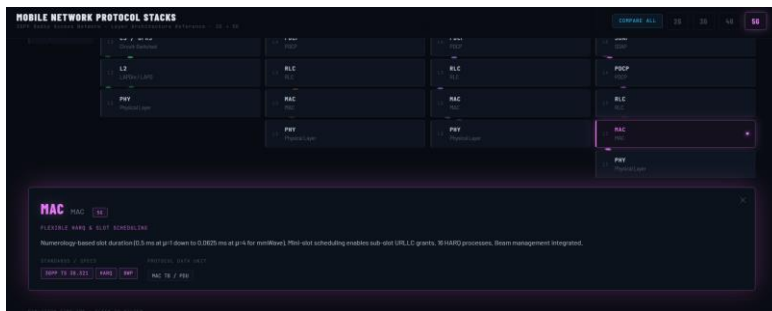
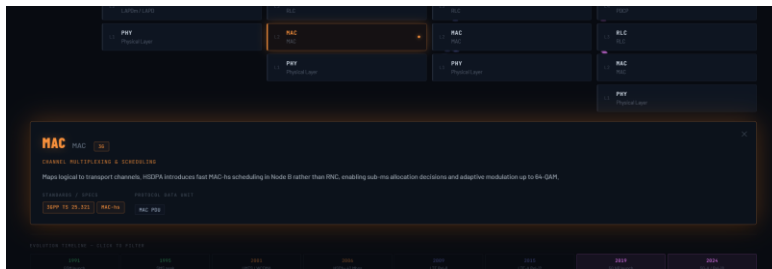
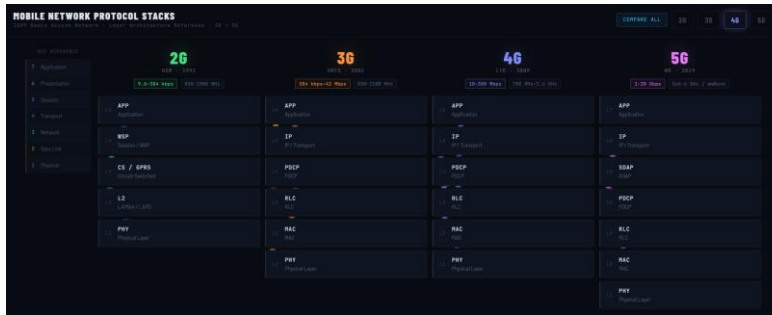


3. <https://www.figma.com/proto/fw6TOXchJcX7bF0pIMUk9L/Untitled?t=2Noa2iaSsBplU2fc-0&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&node-id=1-2&starting-point-node-id=1%3A2>

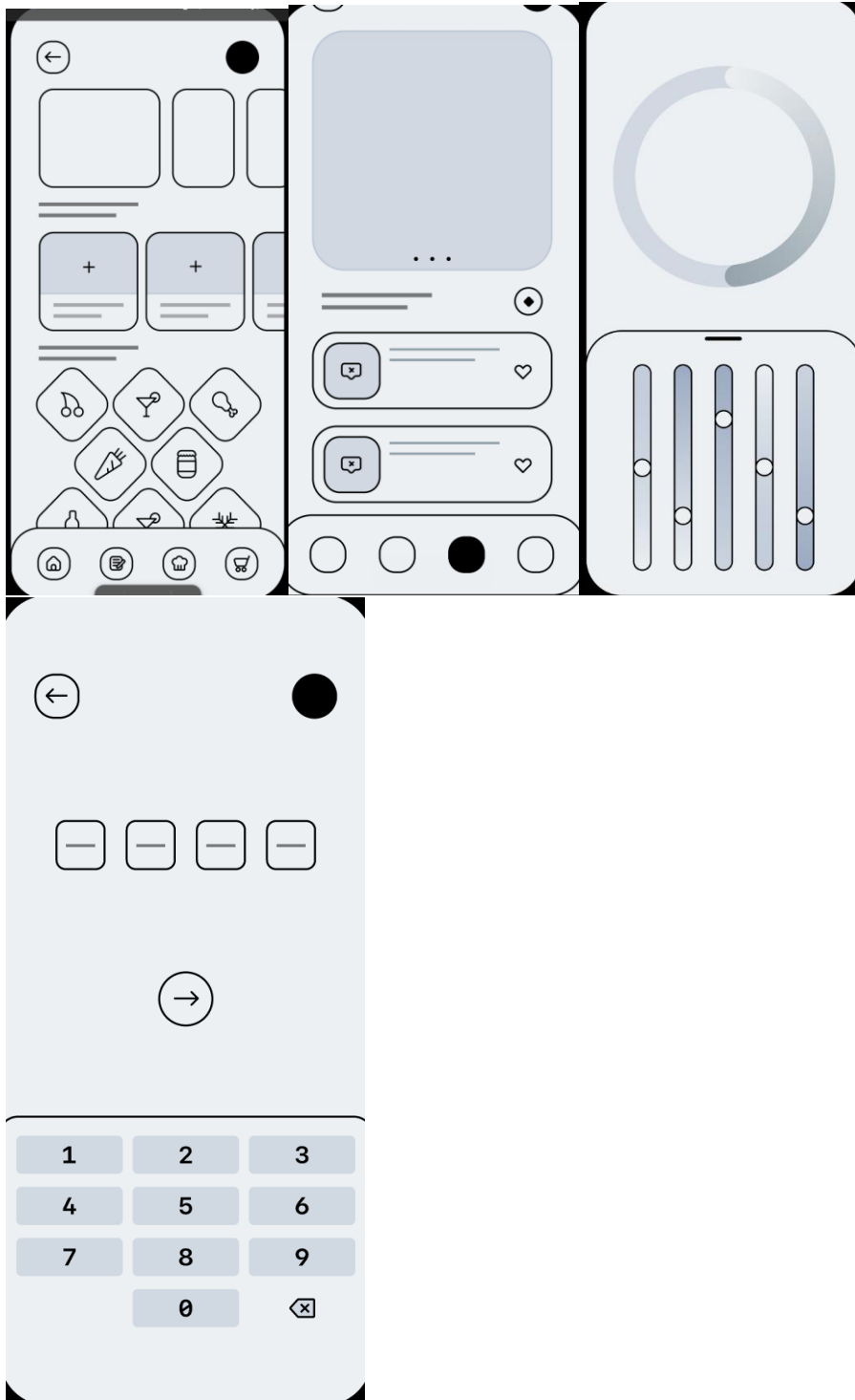
Network Switching Subsystem (NSS) Operation Flowchart



4. figma.com/make/E2TNWX29L6oruEZ3cwue4f/Mobile-Network-Protocol-Stack-Visual?t=75kVvadkCzlvdXQA-20&fullscreen=1&code-node-id=0-6



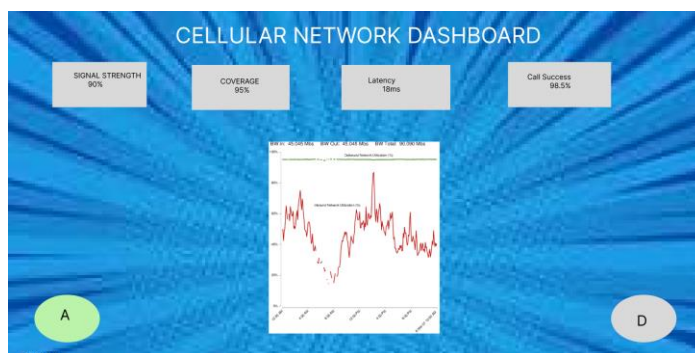
5. <https://www.figma.com/proto/NjMMWONWvHjLLm2YOwBXS2/%F0%9F%93%B2Wireframes-for-mobile-UI-design--Community-?node-id=76-703&p=f&t=7qedpxTi0NsMJYh6-0&scaling=min-zoom&content-scaling=fixed&page-id=35%3A0>

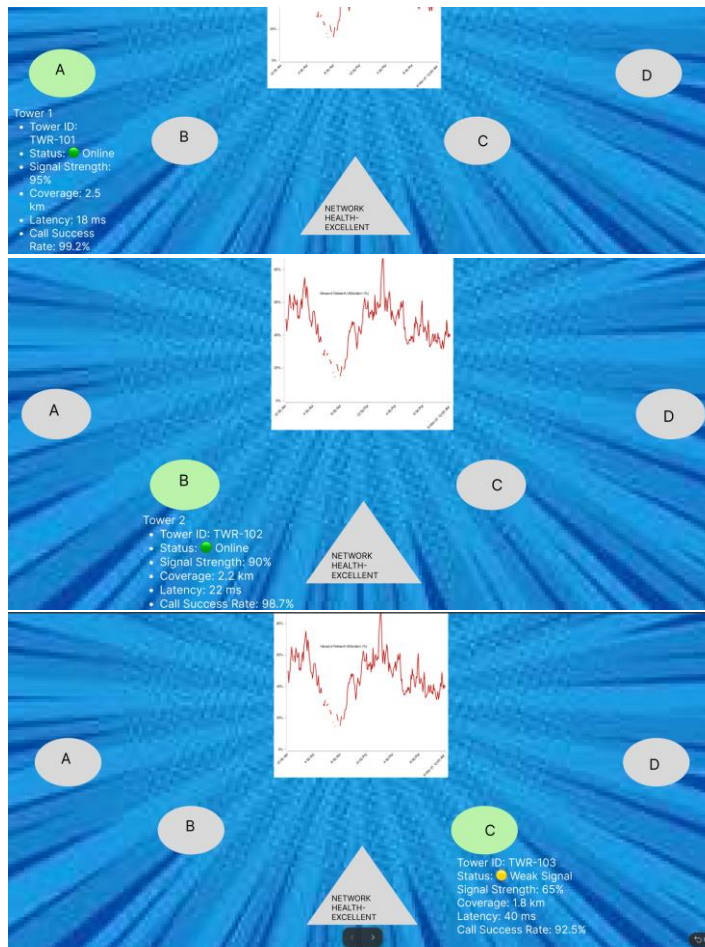


6. <https://www.figma.com/proto/r6wu9GUku9KJ4RbDA1pvLo/Untitled?node-id=1-2&p=f&t=2dldN5oYT1Al44Lx-0&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A2>

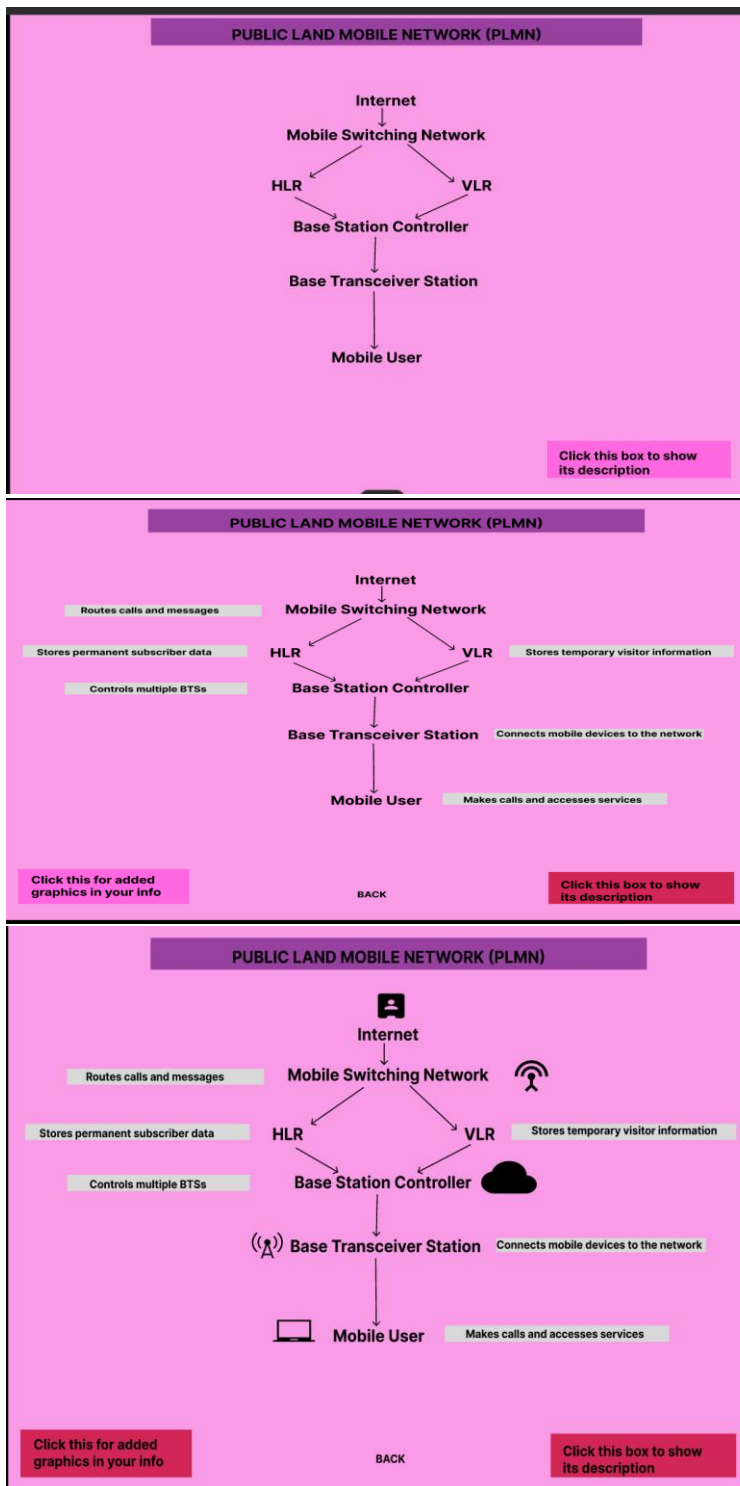


7. <https://www.figma.com/proto/anISFmwv9DLuGFRw8HlrgK/Untitled?node-id=1-32&p=f&t=TikTMmuTj19vfZIY-0&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A3>

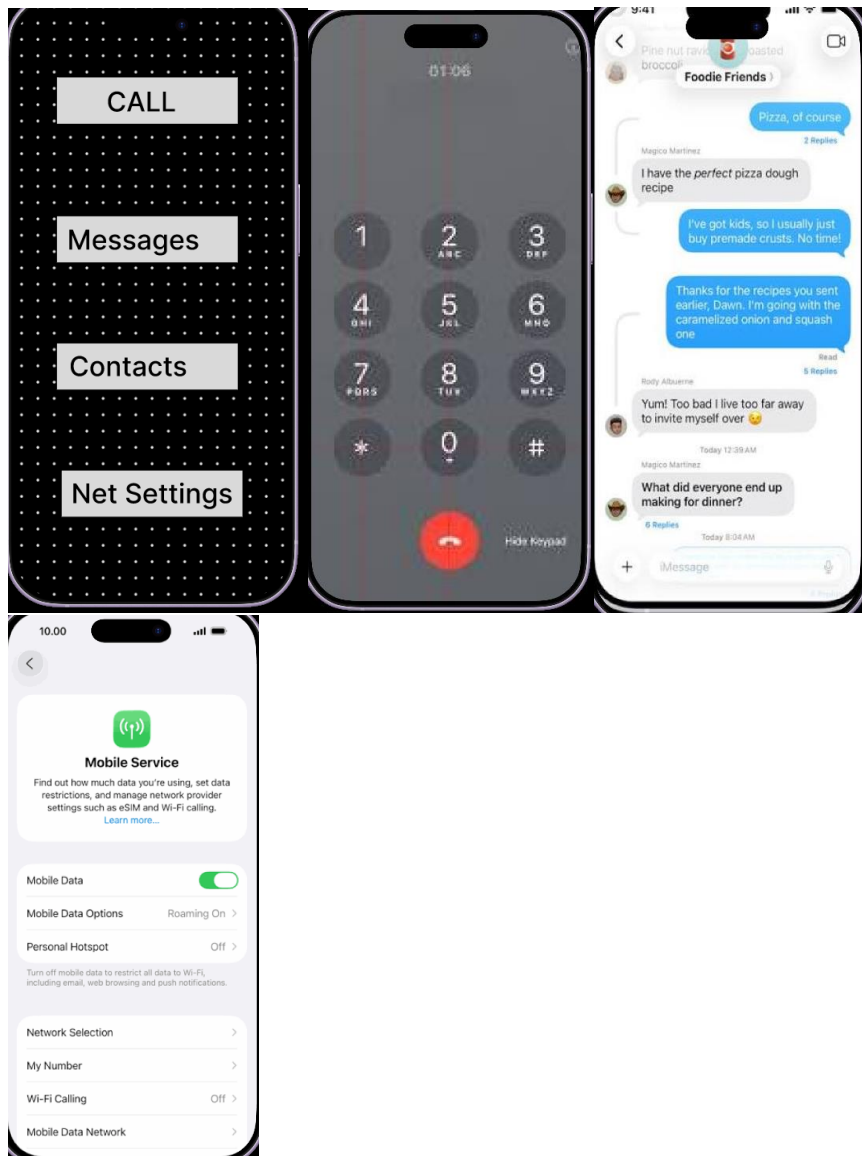




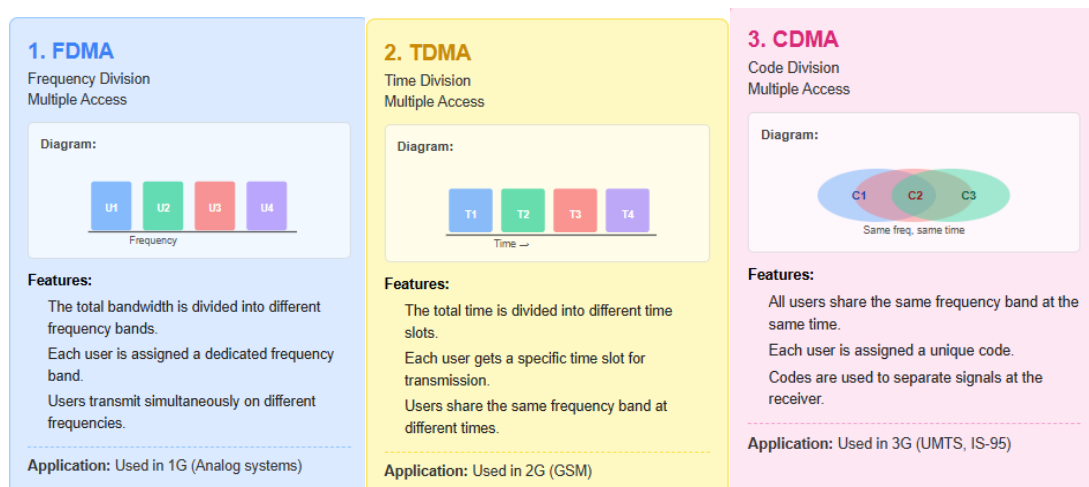
8. <https://www.figma.com/proto/oX5z3a7LdXtAyc8B1bmmvU/Untitled?node-id=1-2&p=f&t=RejCDFTOikF0hutT-0&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A2>



9. <https://www.figma.com/proto/nv9ANwMrysmayYHohZzjf/Untitled?node-id=1-2&p=f&t=RejCDFTOikF0hutT-0&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A2&show-prot-sidebar=1>



10. <https://www.figma.com/make/u3fEfgGxAWwApgAw7digB/Educational-Infographic-Design?code-node-id=0-6&p=f&t=G68SUBweYtVLhCj0-0&fullscreen=1>



4. OFDMA

(Orthogonal Frequency Division Multiple Access)

Diagram:



Features:

- The bandwidth is divided into many orthogonal subcarriers.
- Users are assigned one or more subcarriers.
- Very efficient and supports very high data rate.

Application: Used in 4G (LTE) and 5G

Comparison Summary

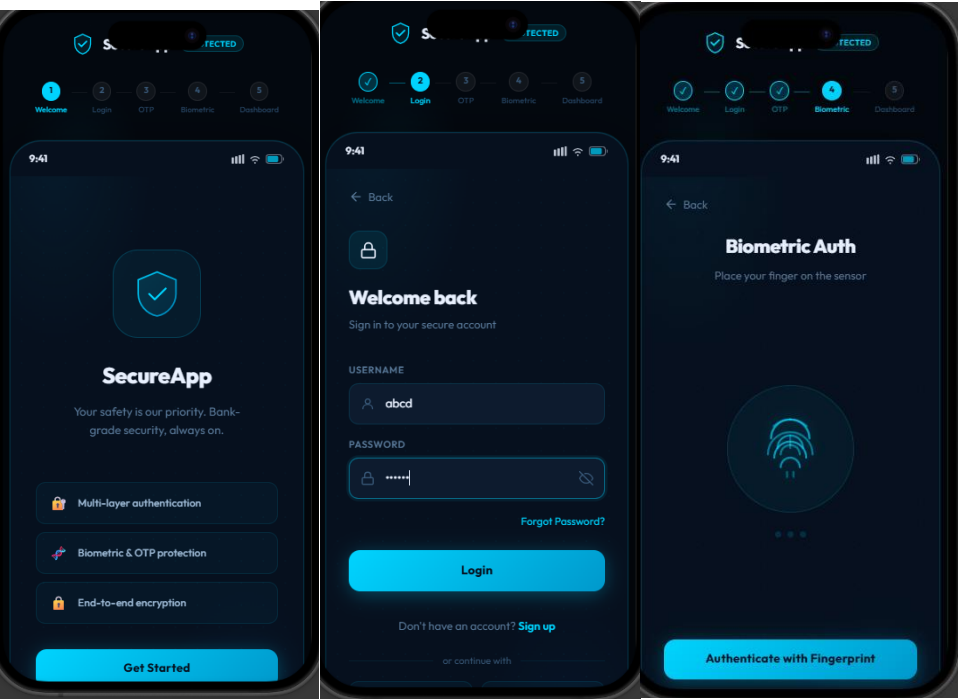
Feature	FDMA	TDMA	CDMA	OFDMA
Access Method	Frequency Division	Time Division	Code Division	Orthogonal Frequency Division
Spectrum Efficiency	Low	Medium	High	Very High
Interference	High (needs guard bands)	Medium	High (self-interference)	Very Low
Synchronization	Not Required	Required	Required	Required
Typical Generation	1G	2G (GSM)	3G (UMTS, IS-95)	4G (LTE), 5G
Main Advantage	Simple to implement	Better spectrum efficiency than FDMA	Capacity and good reliability	High efficiency, very fast data

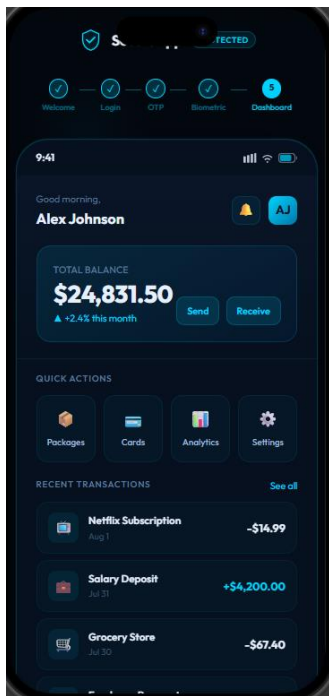
Key Takeaway

The evolution from FDMA → TDMA → CDMA → OFDMA reflects the growing need for higher spectral efficiency and faster data rates. OFDMA (used in LTE and 5G) is the most efficient as it combines the benefits of earlier techniques while minimizing interference.

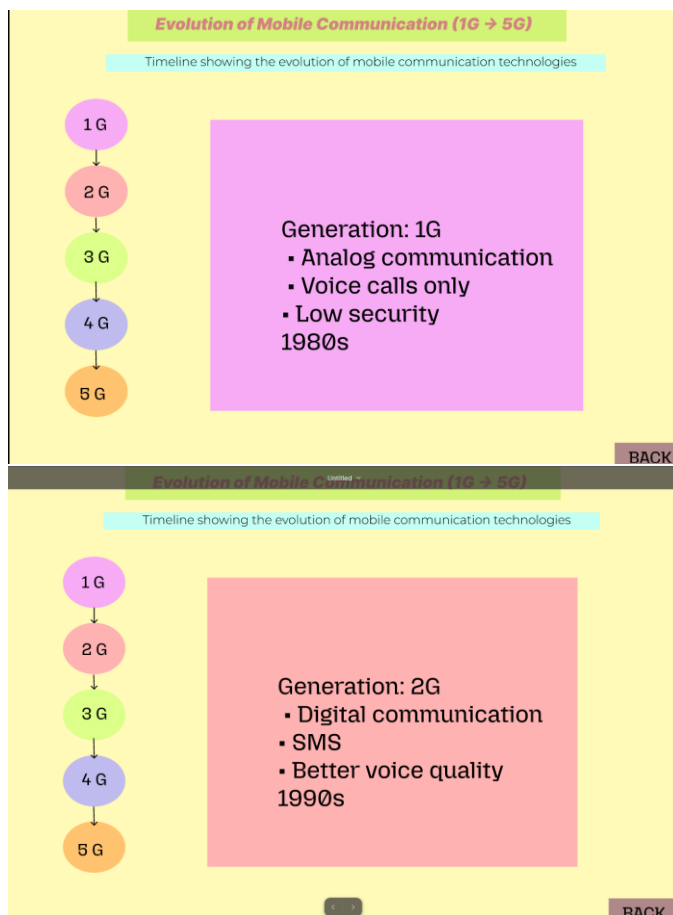
CDMA data users and 4G/5G systems rely on OFDMA for better reliability.

11. <https://www.figma.com/make/lfsjYFL0uT2ZhHFFyGTLGI/Security-flow-wireframe-prototype?code-node-id=0-6&p=f&t=56ATV4RRuid8xPwd-0&fullscreen=1>





12. <https://www.figma.com/proto/GxTSVwOMZ2aD11M788pU5f/Untitled?node-id=1-2&p=f&t=RejCDFTOikF0hutT-0&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A2>





13. <https://www.figma.com/make/XzIW4rjINrKCq2hZLqDhAH/Educational-Infographic-Design?code-node-id=0-6&p=f&t=4zGmnNz9DT8VvcsO-0&fullscreen=1>

Mobile Communication vs Mobile Computing

Mobile Communication	Mobile Computing
Definition Mobile communication refers to the exchange of voice, data or video between mobile devices over a network.	Definition Mobile computing refers to the ability of mobile devices to process, store and retrieve data wirelessly.
Purpose To enable communication and connectivity between mobile users anywhere.	Purpose To provide computing services and information access anytime, anywhere.
Main Technologies GSM, GPRS, EDGE, 3G, 4G (LTE), 5G	Components Mobile Device, Operating System, Applications, Network, Cloud Services
Advantages Wide area coverage Reliable communication Real-time information	Advantages Anytime, anywhere access Increased productivity Supports complex applications
Applications Voice calls, SMS, video calls, mobile internet, IoT communication	Applications Mobile banking, GPS navigation, online learning, mobile office
Examples Making calls, sending SMS, using mobile services	Examples Smartphones, tablets, laptops, Mobile apps, Cloud services

Comparison Table

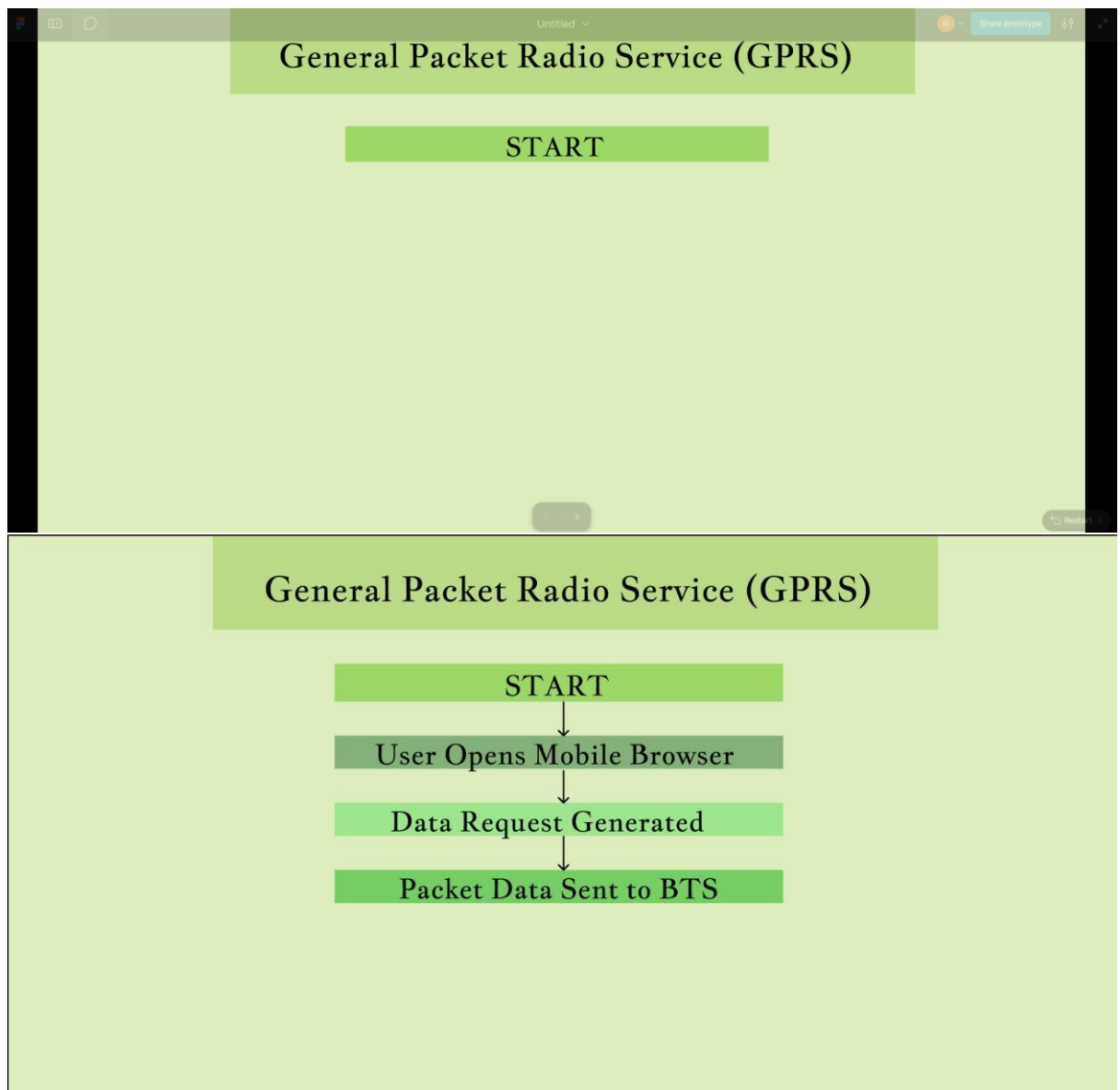
Aspect	Mobile Communication	Mobile Computing
Primary Focus	Exchanging information	Processing information
Connectivity	Network dependent	Network + Local processing
Devices	Basic mobile phones	Smartphones, tablets, laptops
Data Processing	Minimal (mainly transfers)	Done locally or in cloud
Mobility	Enables communication on the move	Enables computing on the move
Typical Examples	Voice call, SMS, Video call	Mobile apps, Cloud services

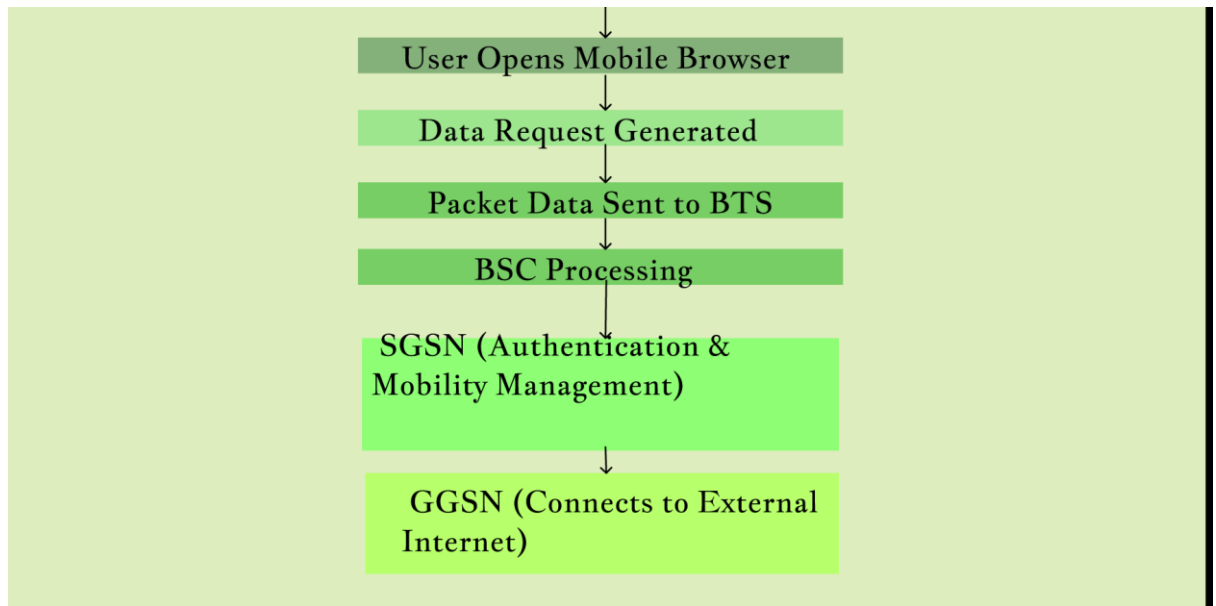
★ **Similarities**

- Both require wireless connectivity.
- Both support mobility and portability.
- Both are essential in modern everyday life.
- Both work together to provide better user experience.
- Both have a wide range of applications.

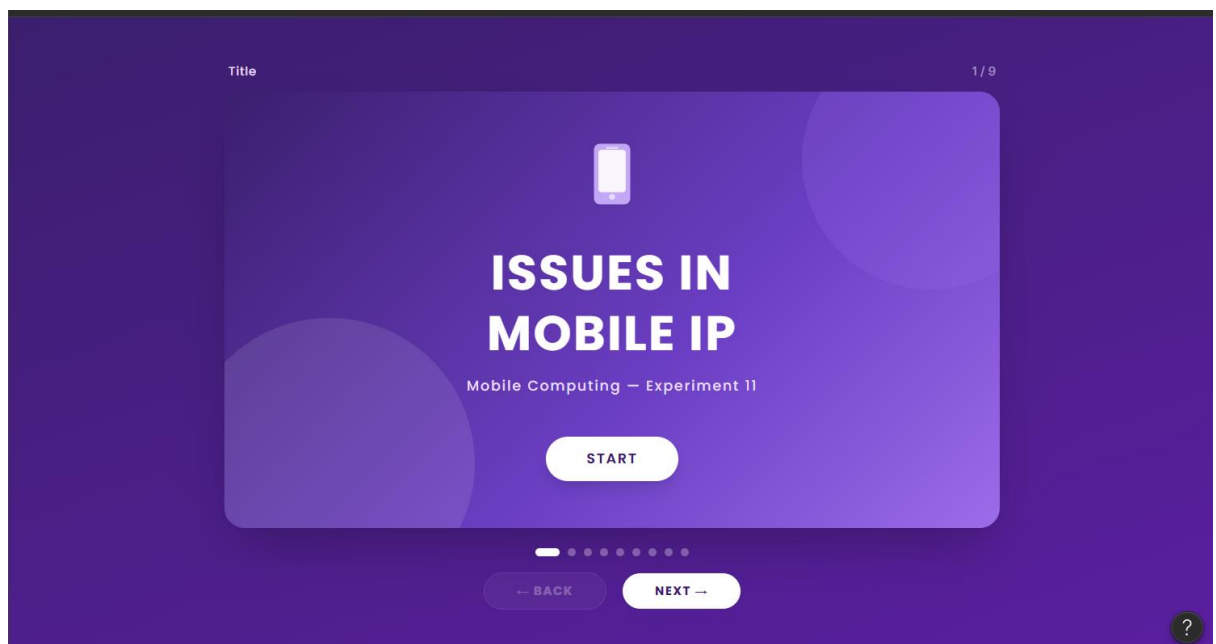
Experiment 13

14. <https://www.figma.com/proto/IeU4qxZmbBN2CcEBXWMICD/Untitled?node-id=1-2&p=f&t=J09AUx1zujs787wY-0&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A2>





15. <https://www.figma.com/make/7Q0DcMvQtsaueJ3JDtTsga/Beginner-friendly-design-creation?t=N7BRQbKxxasSqHYm-1>



What Is Mobile IP?

Mobile IP allows a mobile device to maintain connectivity while moving from one network to another — keeping the same IP address as a home address regardless of location.

**Home Agent**

Router at the home network, forwards packets to mobile node

**Foreign Agent**

Router at the visited network, delivers packets to mobile node

**Mobile Node**

The device roaming between networks



Home

tunnel



Foreign



Mobile

Packets are tunneled from Home Agent to Foreign Agent, then delivered to the Mobile Node.

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NEXT →

2. Handover Delay

2 Handover Delay

When a Mobile Node moves from one network to another, it must register a new Care-of Address with the Home Agent. During this process, packets destined for the Mobile Node may be lost or delayed.

1 Detect new network

2 Acquire Care-of Address

3 Register with Home Agent

4 Resume communication



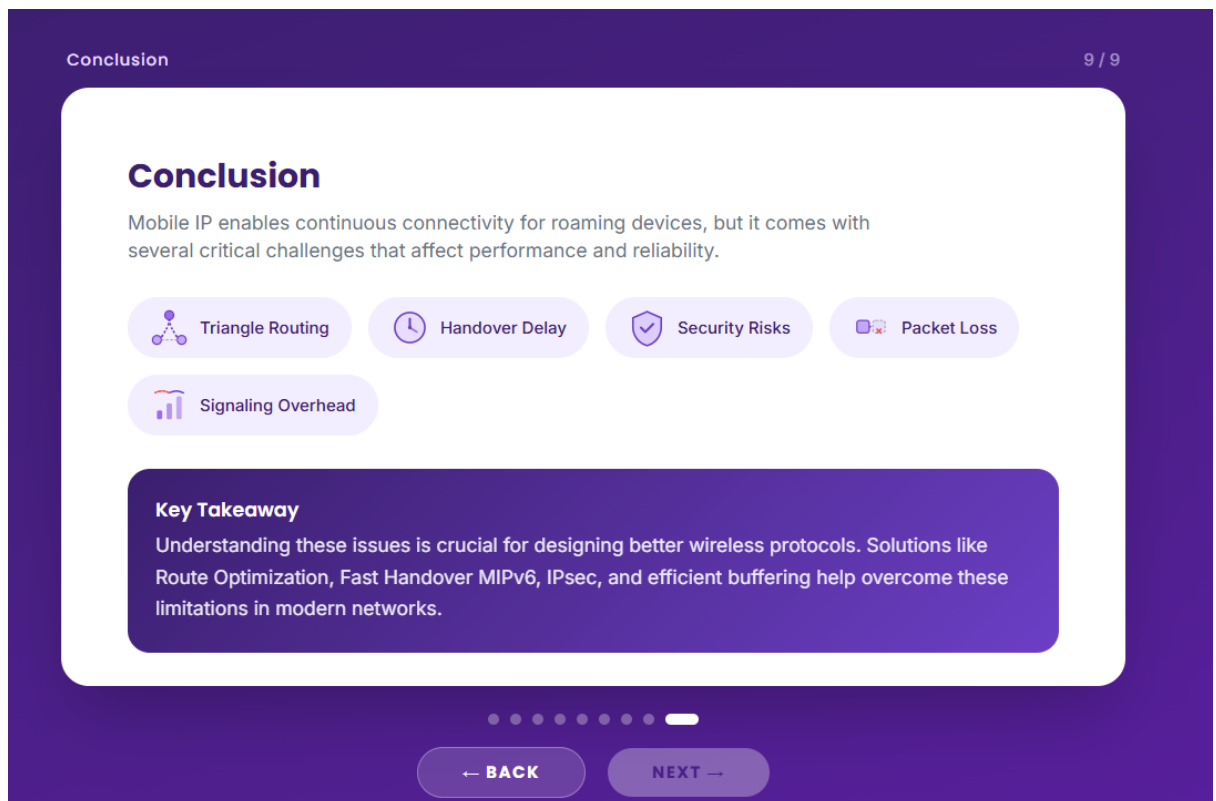
Typical Handover Delay

50–300ms

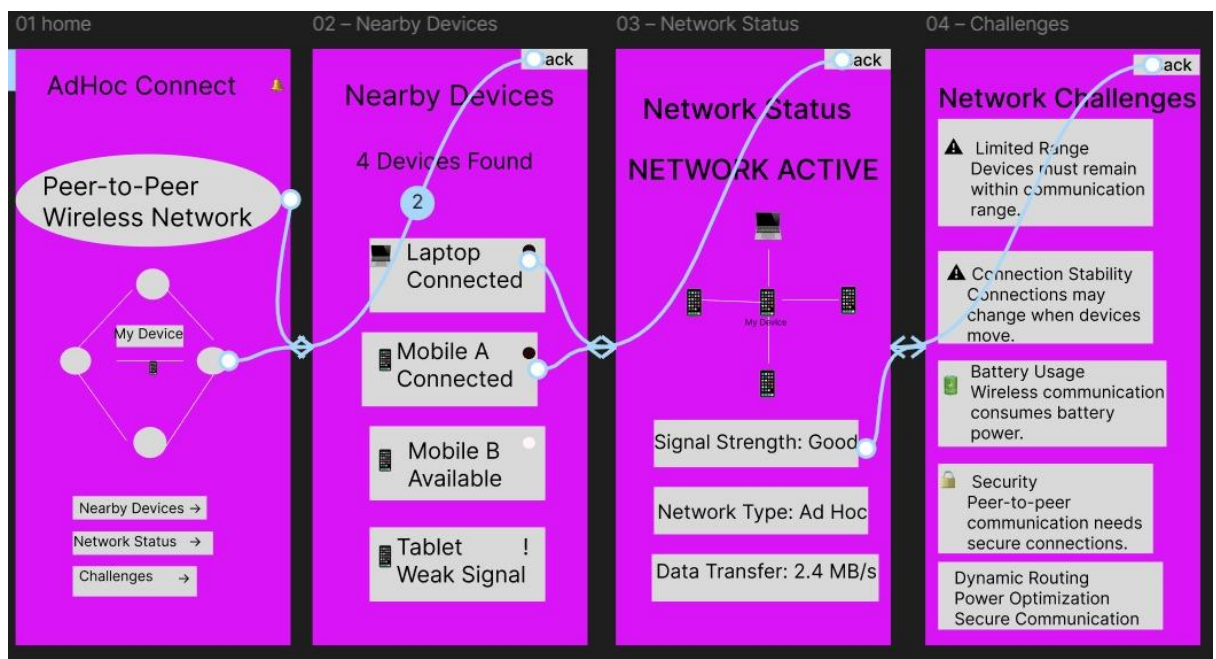
Noticeable in real-time applications like VoIP

← BACK

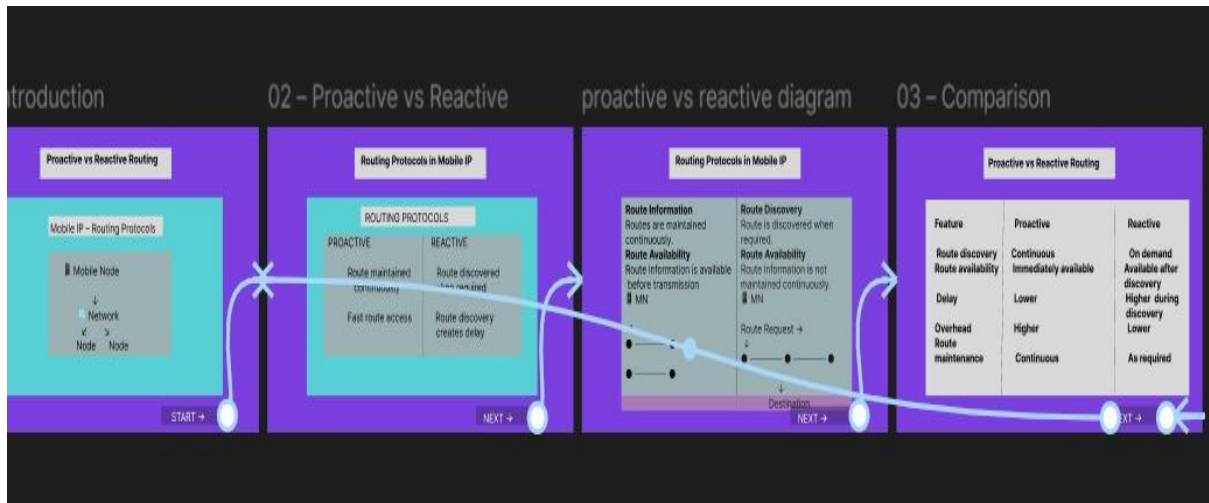
NEXT →



16. <https://www.figma.com/design/xfeolWY6sUNxXiSLIKBdn8/Untitled?node-id=1-7&t=N7BRQbKxxasSqHYm-1>



17. <https://www.figma.com/design/4TOusnCxnGgIKz7ypIH5qC/Untitled?node-id=0-1&t=Rg23c5EVvHtATlbe-1>



18. <https://www.figma.com/design/4TOusnCxnGgIkz7ypIH5qC/Untitled?node-id=4-100&t=Rg23c5EVvHtATibe-1>

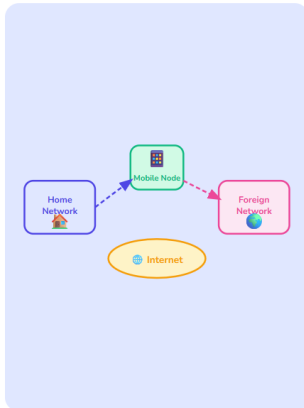


19. <https://www.figma.com/make/T4K2fsPOZE4ELBMHlegOnB/Beginner-friendly-app-builder?code-node-id=0-6&fullscreen=1>



STEP 1 — UNDERSTANDING THE BASICS

What is Mobile IP?



Mobile IP lets your phone or tablet stay connected to the internet even when you move from one network to another — like moving from your home Wi-Fi to a coffee shop network. Think of it like the post office forwarding your mail to your new address!

KEY CONCEPTS

- Mobile Node (MN) — your phone or laptop that moves around
- Home Network — where your device was originally registered
- Foreign Network — the new network you connect to
- Care-of Address — your temporary address at the foreign network

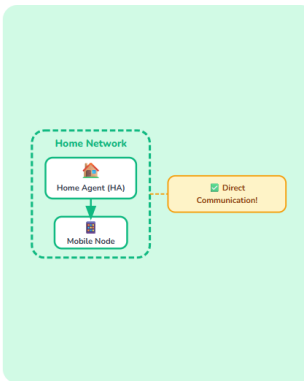
← Previous

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STEP 2 — WHERE IT ALL BEGINS

Your Home Network



The Home Network is like your permanent home address. Even when you travel, your home agent (HA) stays there and manages your mail (data packets). It knows where to forward your messages when you're away.

KEY CONCEPTS

- Home Agent (HA) — the manager that stays at your home network
- Home Address — your permanent IP address (like your home address)
- HA stores all routing info for your device
- Direct communication happens when you're at home

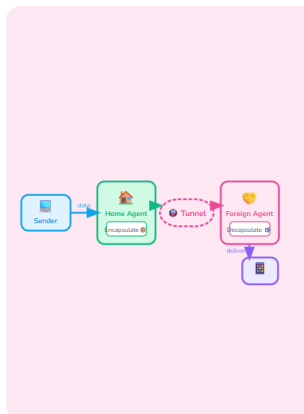
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STEP 5 — HOW YOUR DATA TRAVELS

Sending Data (Tunneling)



Data is sent through a "tunnel" — imagine wrapping a letter inside another envelope. The Home Agent wraps (encapsulates) your data packets and sends them through a tunnel to your Care-of Address. The Foreign Agent unwraps them and delivers them to you!

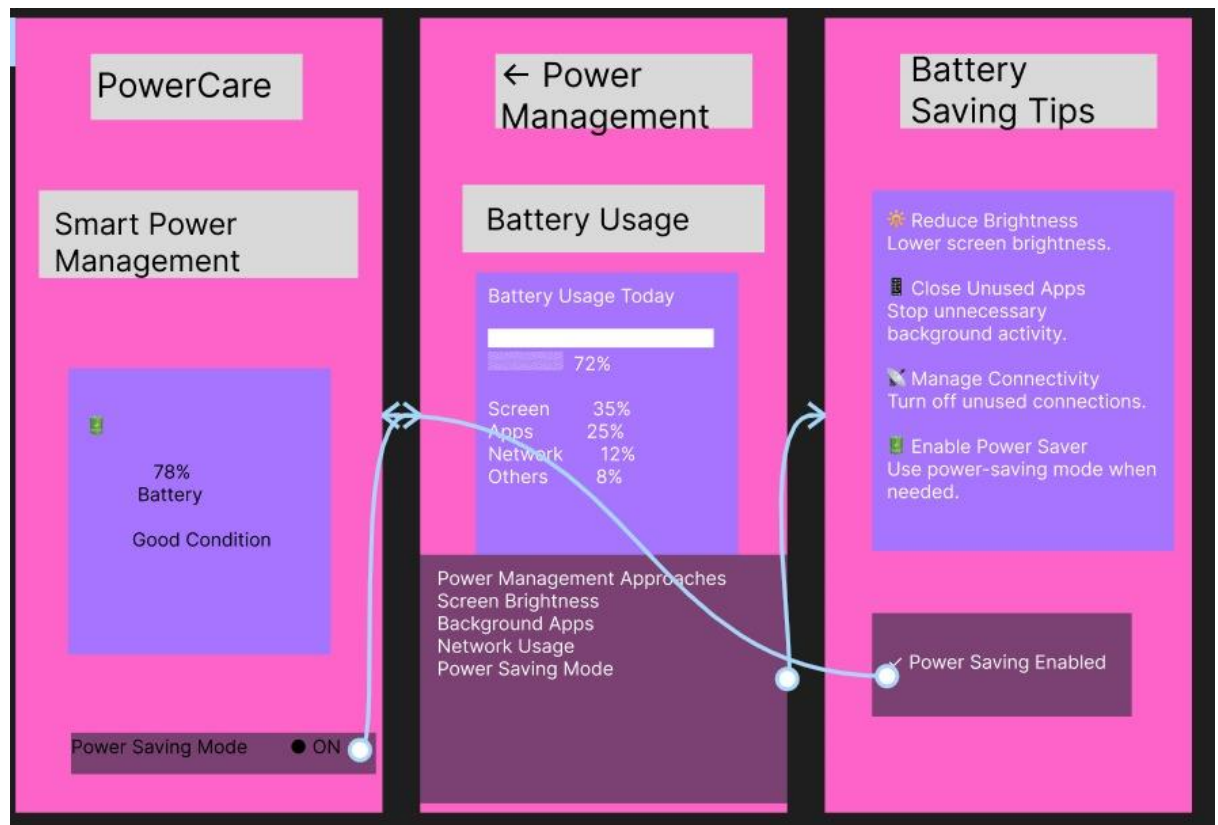
KEY CONCEPTS

- Sender sends data to your permanent Home Address
- Home Agent wraps packets and tunnels them to your CoA
- Foreign Agent unwraps and delivers packets to you
- Your replies go directly to the sender (triangle routing)

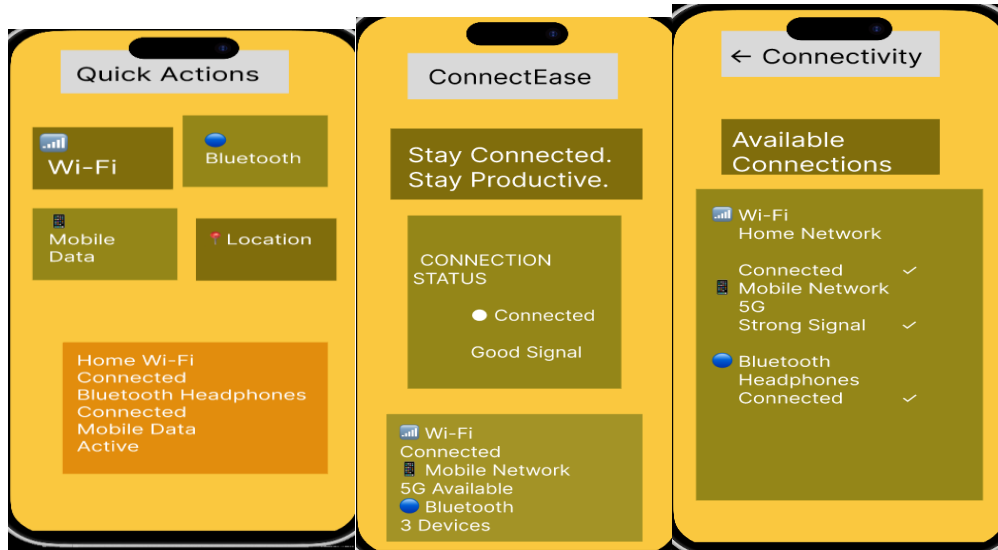
← Previous

Start Over

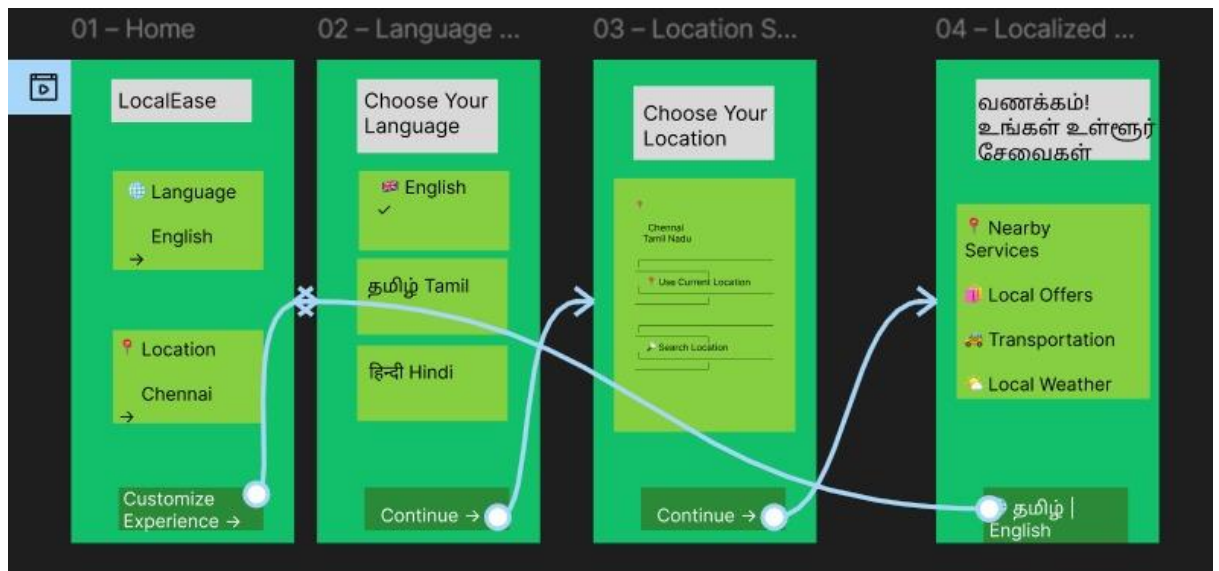
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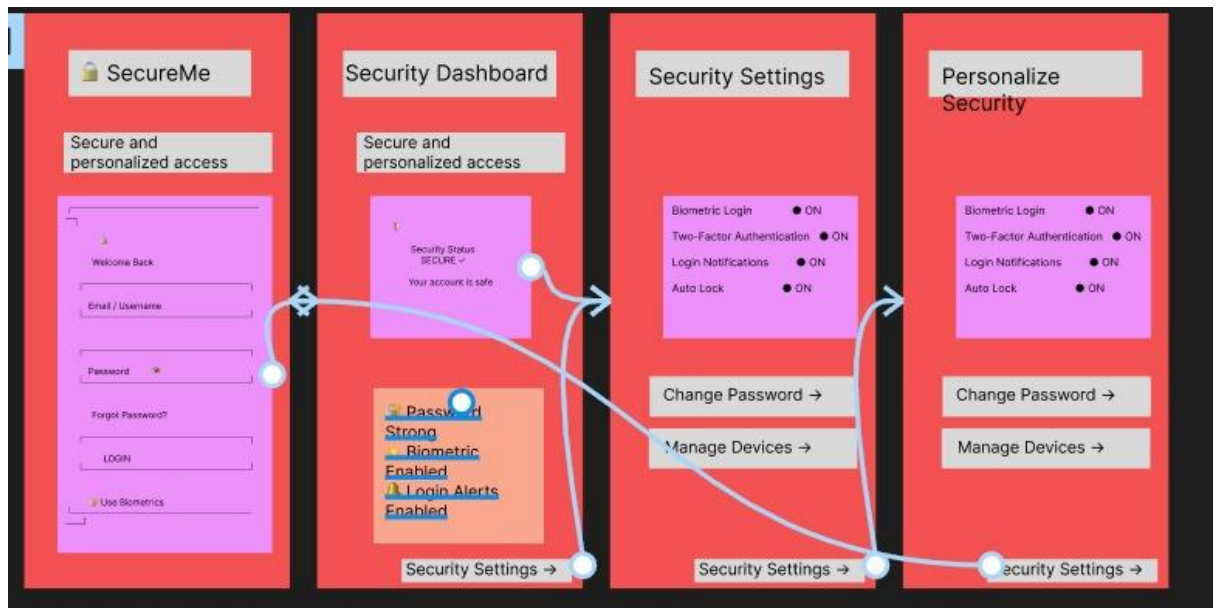
21. <https://www.figma.com/design/6iiSayxhG59oqtgOHZv2o0/Untitled?node-id=1-58&t=Rg23c5EVvHtATlbe-1>



22. <https://www.figma.com/design/6iiSayxhG59oqtgOHZv2o0/Untitled?node-id=2-29&t=Rg23c5EVvHtATlbe-1>



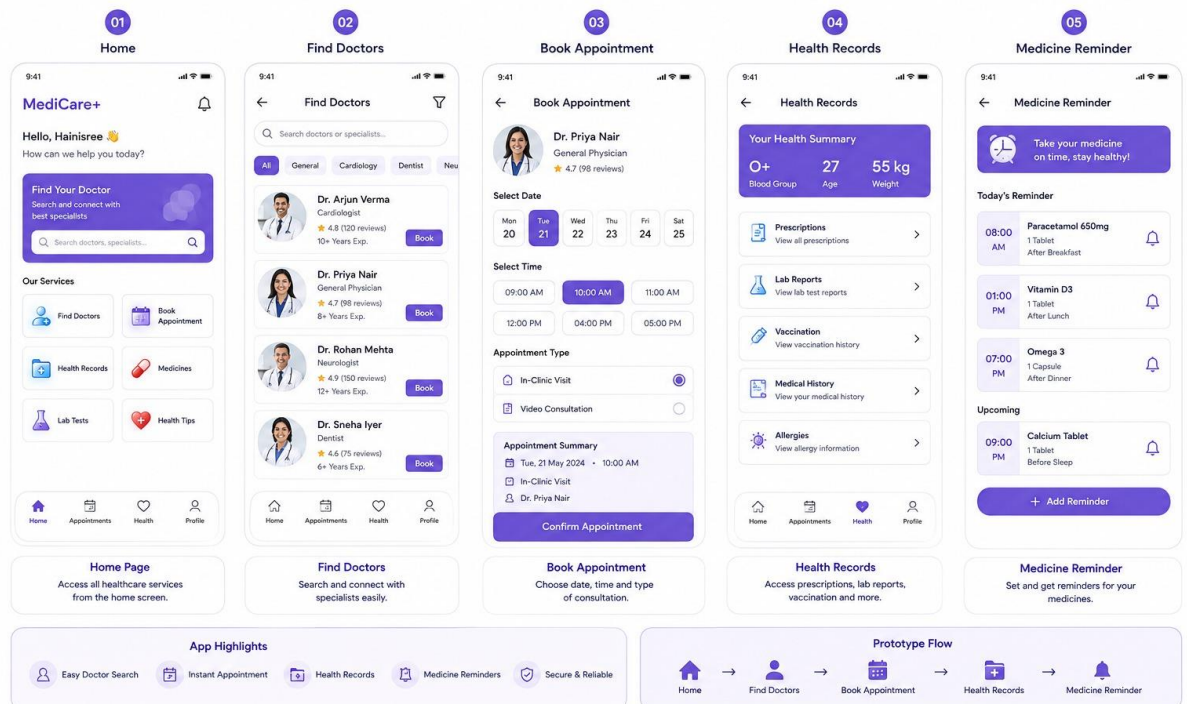
23. <https://www.figma.com/design/6iiSayxhG59oqtgOHZv2o0/Untitled?node-id=7-97&t=Rg23c5EVvHtATlbe-1>



24. <https://www.figma.com/make/99OeuKF9r2XtYn0U92vb5c/Beginner-friendly-Prototype-Design?t=2HLaWAiz1XSypDBK-1>

Experiment 24 – Healthcare Support App

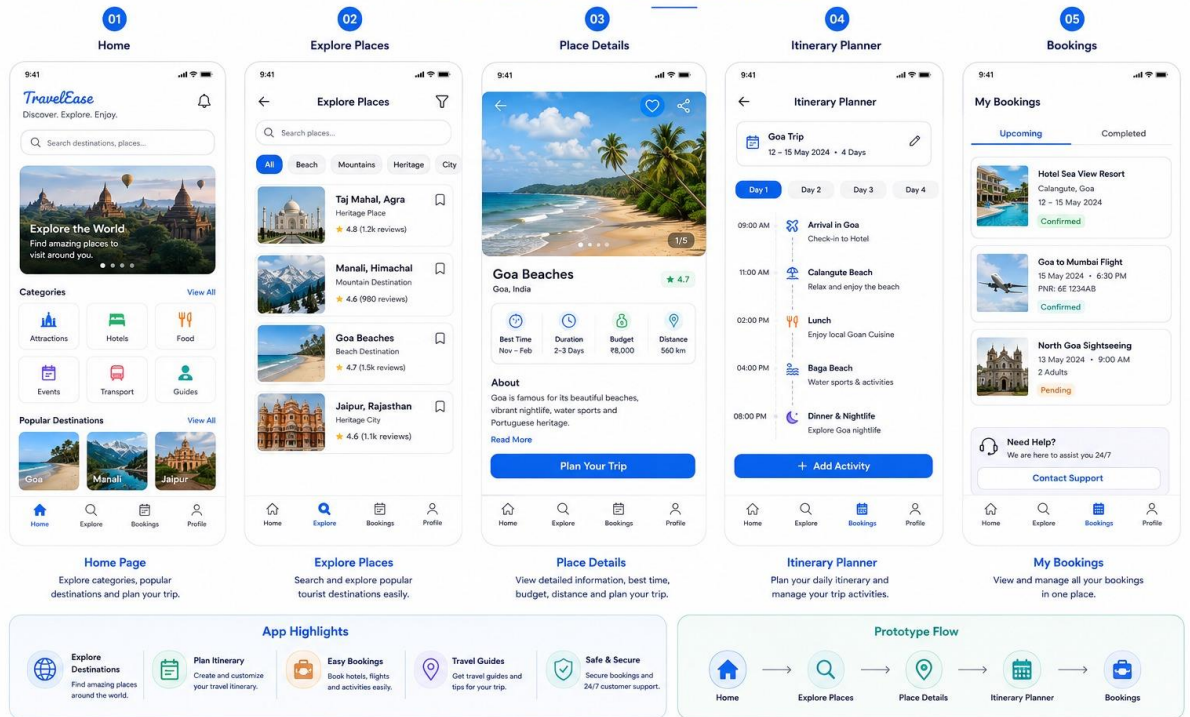
Figma Prototype – Healthcare Services for Medical Support



25. <https://www.figma.com/make/txgz6cLgDB7KEFGy4aQ4ZU/Beginner-Prototype-Creation?t=2HLaWAiz1XSypDBK-1>

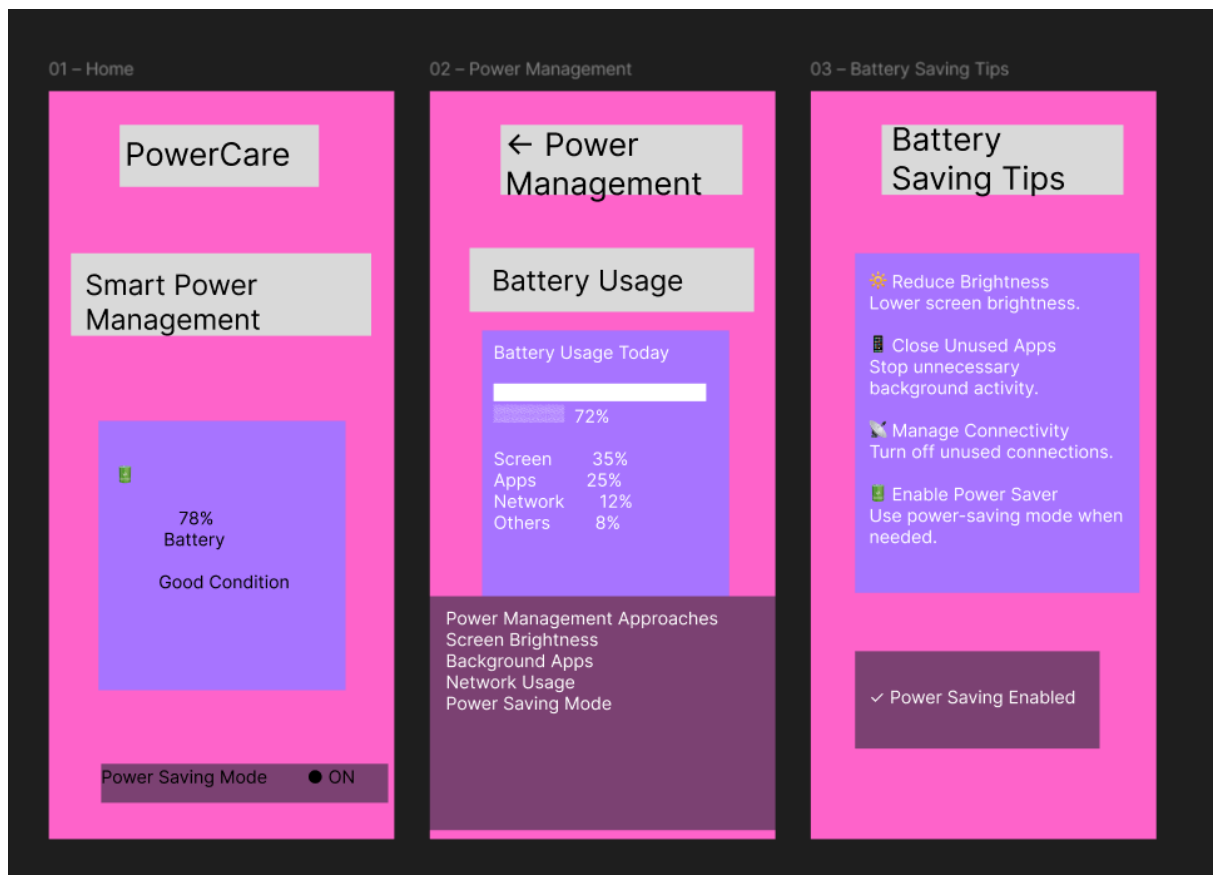
EXPERIMENT 25 – SMART TOURISM APP

Figma Prototype – Smart Tourism Mobile Application



26. <https://www.figma.com/design/wQolCNFy3A90Fzdfm2SkEo/Untitled?node-id=0-1&t=2HLaWAiz1XSypDBK-1>
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29. <https://www.figma.com/design/6iiSayxhG59oqtgOHZv2o0/Untitled?node-id=0-1&p=f&t=NXlQI36UfpS8aCPI-0>



30. <https://www.figma.com/proto/4PpklmxLHb4cDzUSyNzyl/Music-App-UI-Kit--Community-?node-id=40-4801&p=f&t=LmDVGeiRmOCvS3WW-0&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=40%3A4801>

